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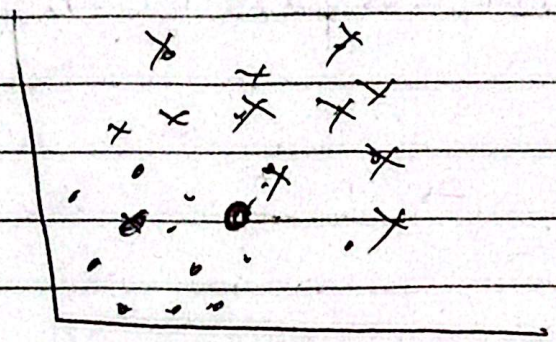
KNN [K Nearest Neighbors].



- calculates distance from each points.
- then sort in ascending order.

Let $k=3$

- Then, it takes 3 nearest points from O
- Majority count [voting]
  - ~~Highe~~ The result which is in majority

There : 2 .

1 X

Hence, O will be .

- It works in every dimension.
- Do Feature Scaling and do transformation [Standard Scaling].

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from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifiers (n_neighbors = 5)

Hyper parameter → k = ?

- How to find best value of K?

a) Heustric (अनुमान)

$k = \sqrt{n}$ ← no. of observations (rows)

and also, avoid even values.

b) Experimentation (try diff value). Better

Cross - Validation.

~~use~~ Check accuracy and find best.

Use loop.

& compare by graph.

plt.plot(range(1,16), scores)

Score = []

loop.

score.append (accuracy_score - - -)

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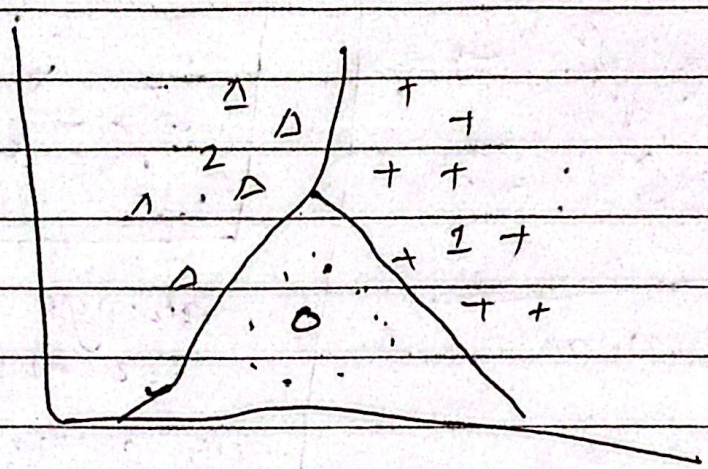
Decision Surface / Boundary

↳ hyper surface that partitions the underlying space into two or more sets, one for each class.



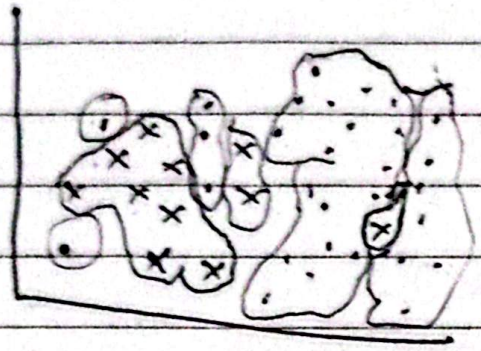
- Voronoi diagram
  - It is a partitioning of a plane into regions based on distance to points in a specific subset of the plane.

mlxtend. → to create decion boundary.

Overfitting & Underfitting in KNN

Overfitting:

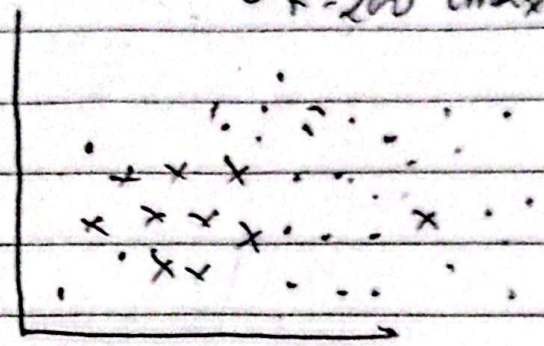
K = 1.



high variance.

Under fitting:

K = 200 [max]



always •

(or most of all are •

Here, compare with all and ~~first~~ the most and highest will be •.

Limitation of KNN

- Lazy & slow

  (or all happen in ~~testing~~ prediction phase, not in training.

  after get query point.

- High dimensional data

  ↳ curse of dimensionality.

  ↓

  can't calculate the distance properly.

- Can't work better with outliers.

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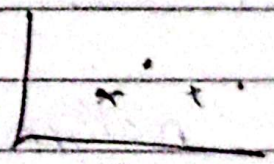
- Imbalanced data..
  - Yes : 98%
  - No : 2%

  } biased towards yes

- Inference and ~~not~~ better for prediction.
  - not good in
  - can't say how much contribution of the feature in the output



Cgpacr sq more contribution?